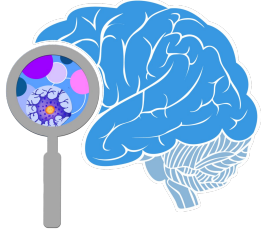


Blood

Lesson objectives:

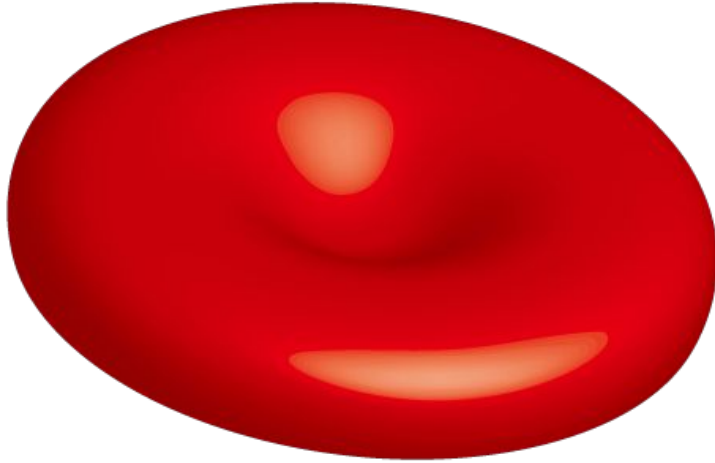
- To describe the components of blood and their functions.



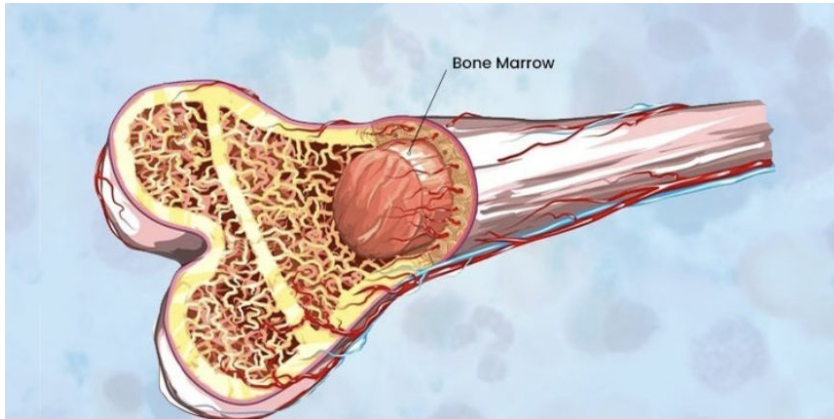
In Year 7 you learnt about **specialised cells**.
What do you remember about **red blood cells**?
What is their **function**?
How is their **structure** adapted to the function?



Red blood cells

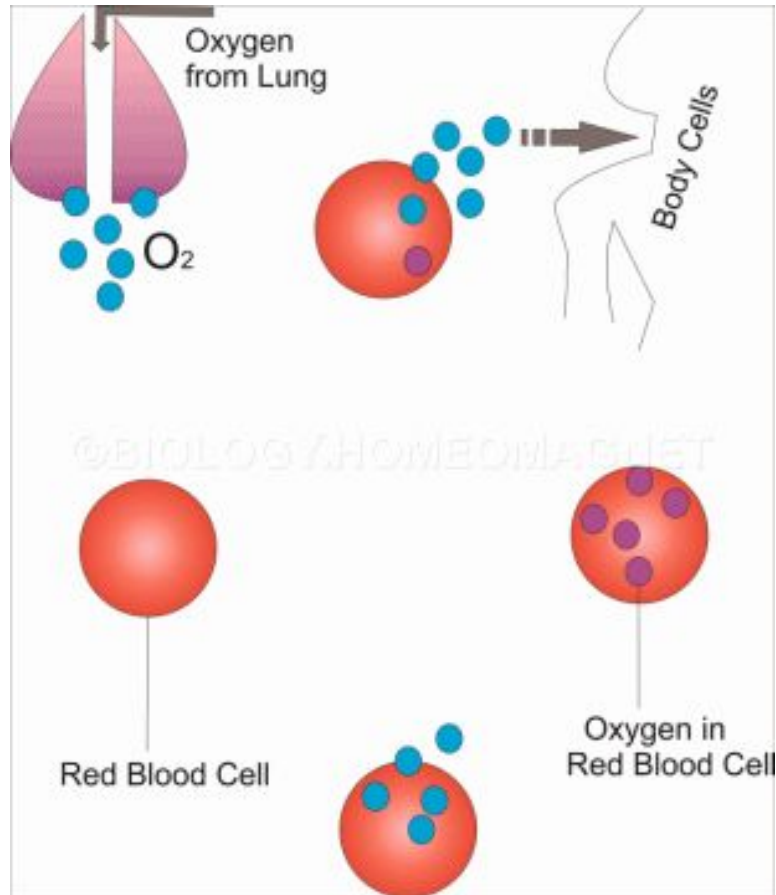


- Red blood cells have dips on both sides to **maximise their surface area to absorb oxygen** as quickly as possible. This shape is called **biconcave**.



- They **do not have a nucleus** to **maximise the oxygen** they can carry.
- Produced in the **bone marrow**.

Red blood cells



- Oxygen moves by **diffusion** into red blood cells in the lungs.
- They have a chemical inside them called **haemoglobin**, a red pigment which binds with the oxygen to carry oxygen to where it is needed for respiration.
- This reaction is reversed near the cells.



Quiz time!

Where are red blood cells made?

A Heart

B Liver

C Bone marrow

D Brain

Quiz time!

Where are red blood cells made?

A Heart

B Liver

C Bone marrow

D Brain

C

Quiz time!

Which cell component are red blood cells missing?

- A** Mitochondria
- B** Cytoplasm
- C** Cell membrane
- D** Nucleus

Quiz time!

Which cell component are red blood cells missing?

- A Mitochondria
- B Cytoplasm
- C Cell membrane
- D Nucleus

D

Quiz time!

What moves into the blood in the lungs?

A Nitrogen

B Water

C Oxygen

D Carbon dioxide

Quiz time!

What moves into the blood in the lungs?

A Nitrogen

B Water

C Oxygen

D Carbon dioxide

C

Quiz time!

By which process does oxygen move from your red blood cells into cells?

A Respiration

B Diffusion

C Active transport

D Osmosis

Quiz time!

By which process does oxygen move from your red blood cells into cells?

A Respiration

B Diffusion

C Active transport

D Osmosis

B

Quiz time!

Which process in your cells requires oxygen to release energy?

A Photosynthesis

B Respiration

C Digestion

D Protein synthesis

Quiz time!

Which process in your cells requires oxygen to release energy?

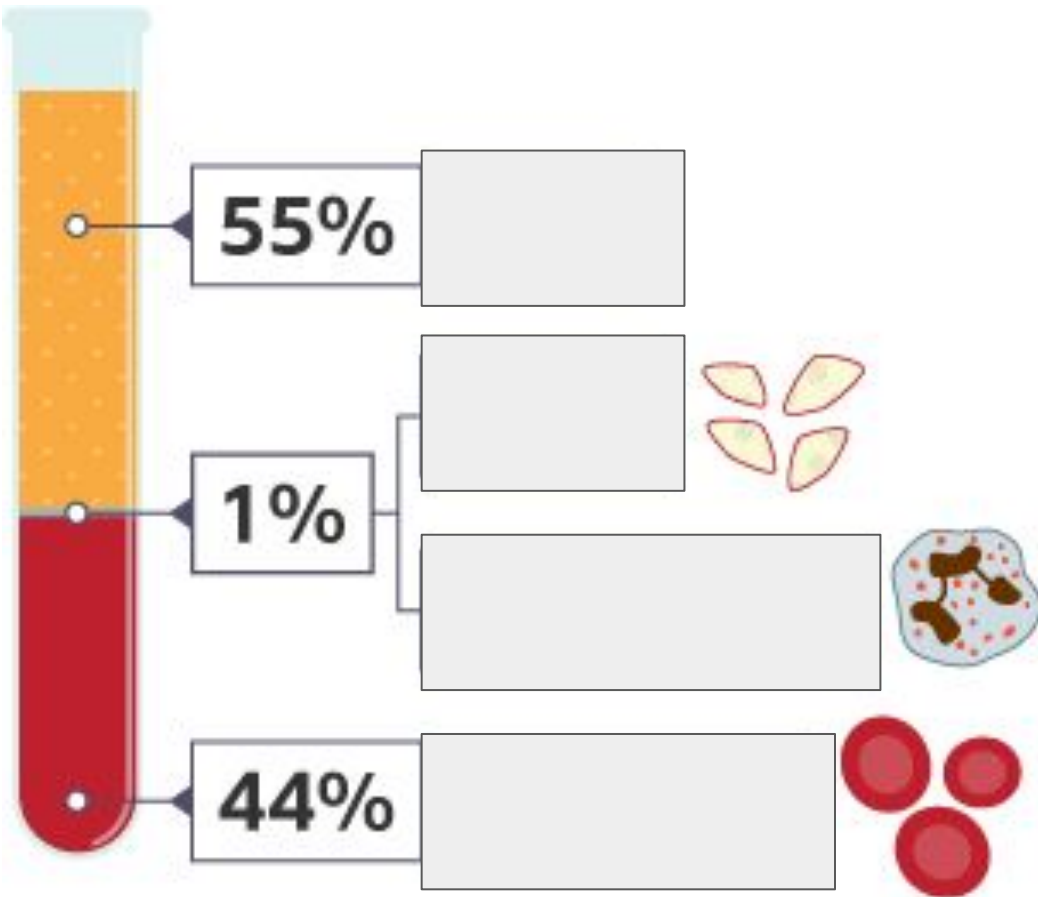
A Photosynthesis

B Respiration

C Digestion

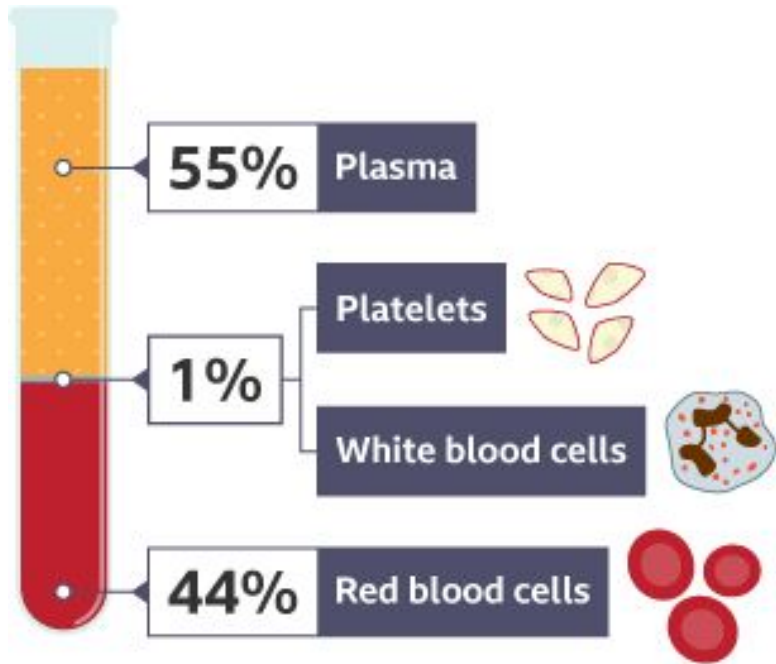
D Protein synthesis

B



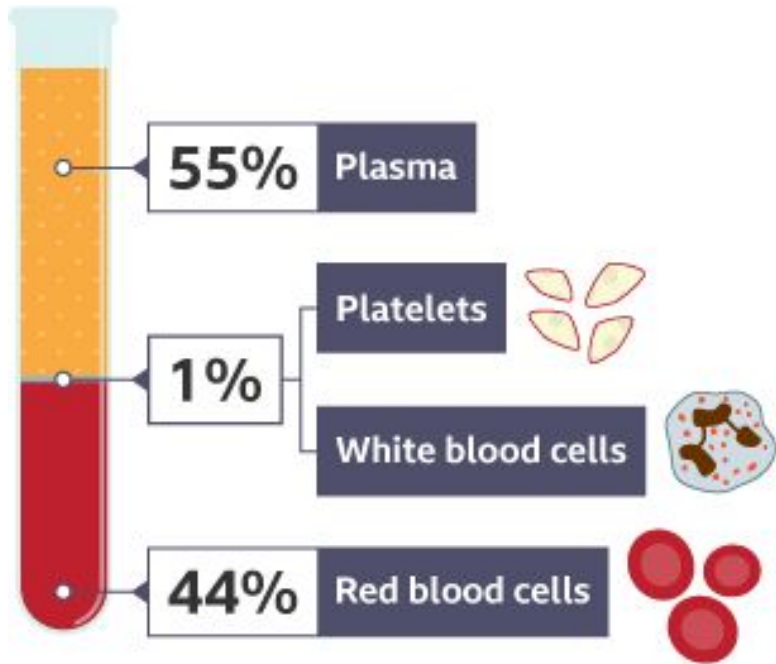
Your blood consists of 4 components, do you know what they are?

Components of blood



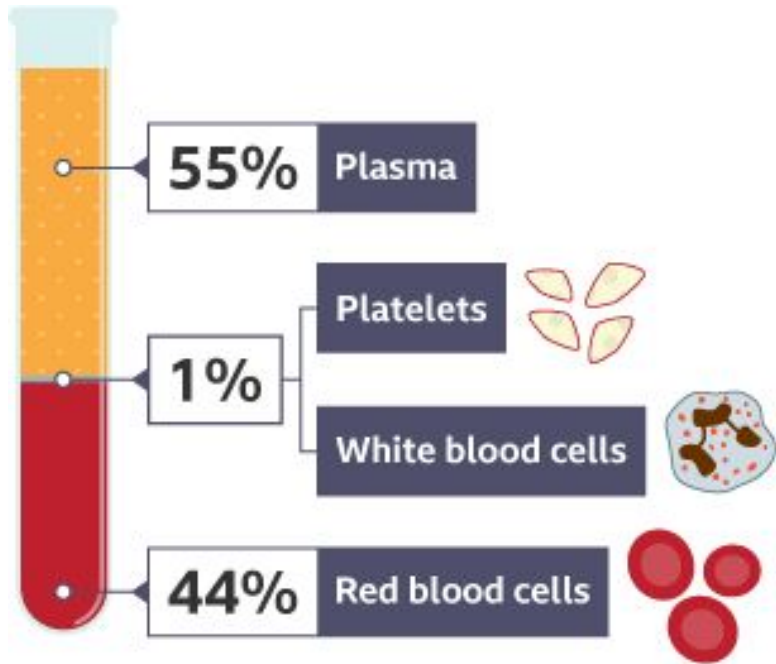
Component	Function
Plasma	
Platelets	
White blood cell	
Red blood cell	

Components of blood



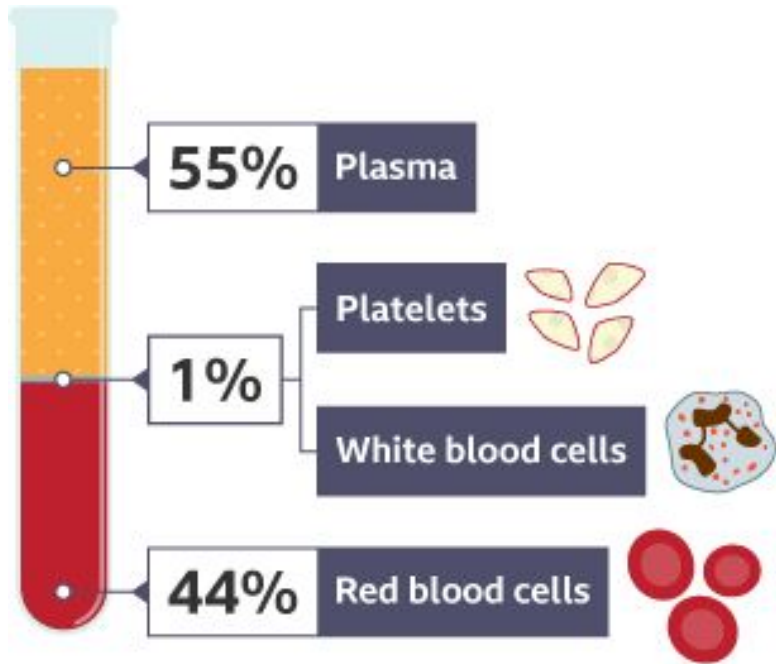
Component	Function
Plasma	Transport blood cells and other dissolved substances around the body e.g. glucose, carbon dioxide
Platelets	
White blood cell	
Red blood cell	

Components of blood



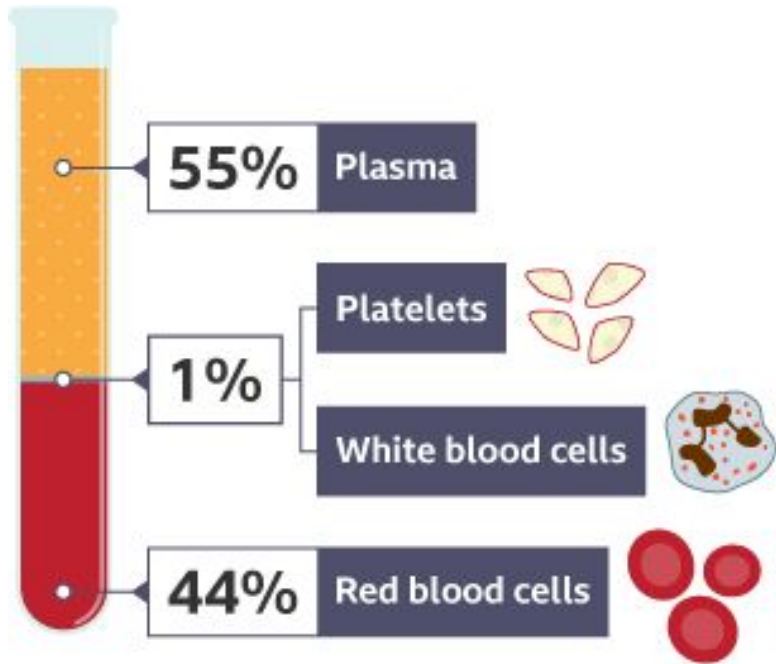
Component	Function
Plasma	Transport blood cells and other dissolved substances around the body e.g. glucose, carbon dioxide
Platelets	Form scabs
White blood cell	
Red blood cell	

Components of blood

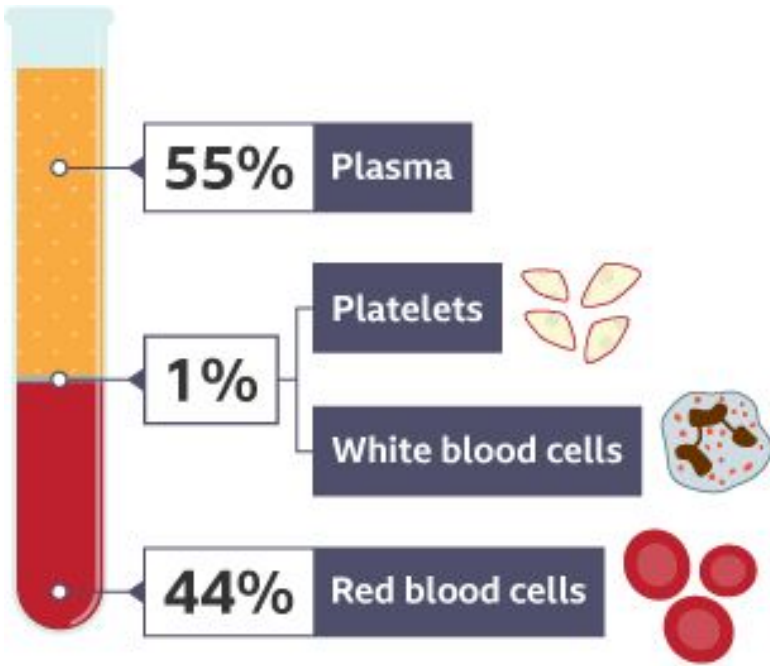


Component	Function
Plasma	Transport blood cells and other dissolved substances around the body e.g. glucose, carbon dioxide
Platelets	Form scabs
White blood cell	Fight infection by pathogens and stop disease
Red blood cell	

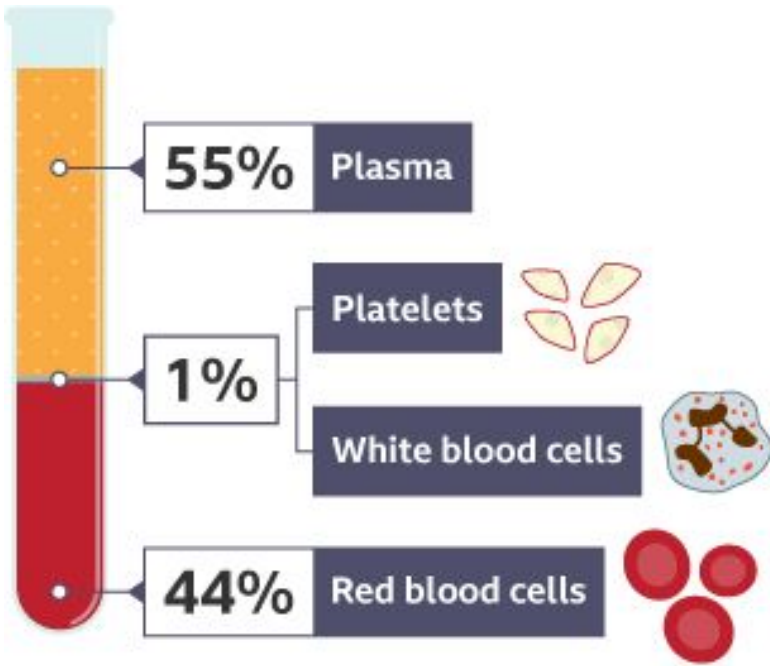
Components of blood



Component	Function
Plasma	Transport blood cells and other dissolved substances around the body e.g. glucose, carbon dioxide
Platelets	Form scabs
White blood cell	Fight infection by pathogens and stop disease
Red blood cell	Carry oxygen from the lungs to every cell in the rest of the body



Component	Function
Plasma	
Platelets	
White blood cell	
Red blood cell	



Component	Function
Plasma	
Platelets	
White blood cell	
Red blood cell	

White blood cells



- White blood cells form part of the **immune system** to keep us safe from infection and disease.
- There are two types of white blood cell:
 - **Phagocytes** which surround, engulf and destroy pathogens
 - **Lymphocytes** which produce antibodies that stick pathogens together. This makes it easier for phagocytes to destroy the pathogens.
- Produced in the **bone marrow**.

Lymphocytes

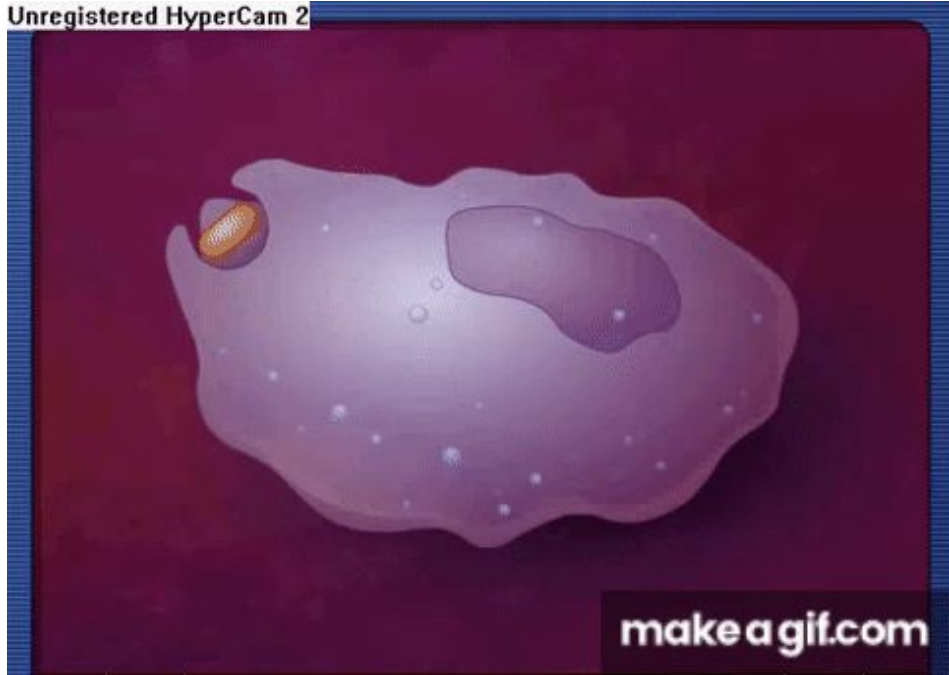


Phagocytes

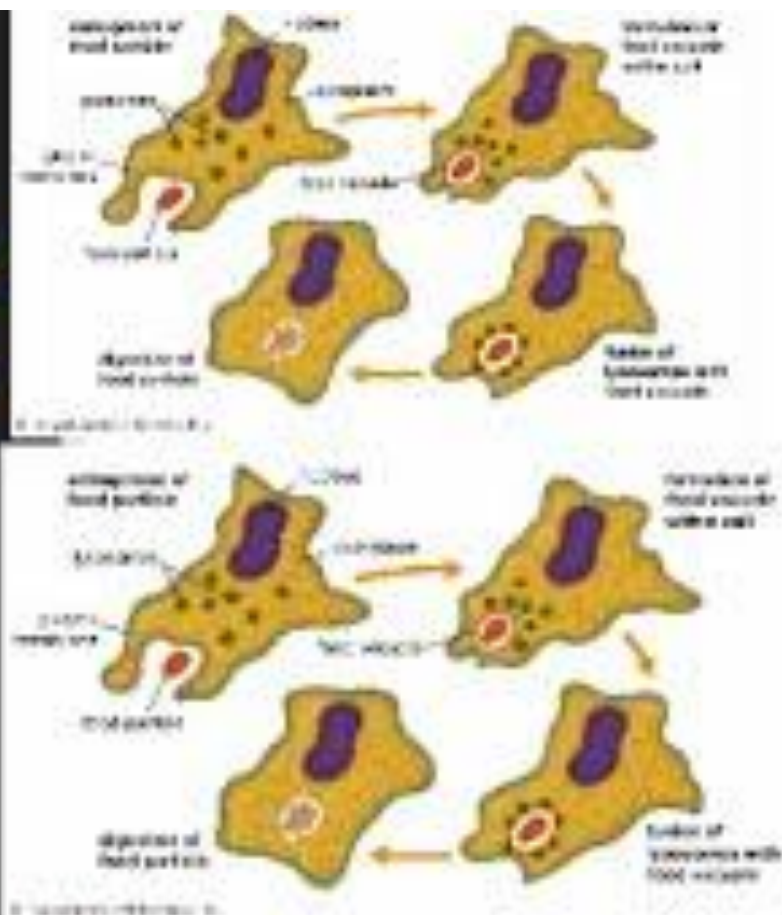
Phagocytes surround, engulf and destroy pathogens



Unregistered HyperCam 2



PHAGOCYTOSIS OF CELL



Phagocytes

COUNTDOWN
10:00
MIN.
with SIMPLE BEEPS



Keywords:

Engulf 吞噬 / 삼키다

Destroy 摧毀/파괴하다

Phagocyte 吞噬細胞

/식세포

Pathogen (eg. bacteria) 病

原体(細菌)/병원체(박테리아)

In small groups, **write a voiceover** for the gif to **describe what happens during phagocytosis.**

Every member of the group should have it written in their books.

Each group will have the opportunity to speak their voiceover.

Challenge:

Can you think of a real-life example where this might happen?

Phagocytes



Quiz time!

Which of these is not a type of white blood cell?

A Lymphocyte

B Phagocyte

C Erythrocyte

Quiz time!

Which of these is not a type of white blood cell?

A Lymphocyte

B Phagocyte

C Erythrocyte

C

Quiz time!

True or false: There are more white blood cells than red blood cells in your body.

A False

B True

Quiz time!

True or false: There are more white blood cells than red blood cells in your body.

A False

B True

A

Quiz time!

True or false: White blood cells are part of your immune system.

A True

B False

Quiz time!

True or false: White blood cells are part of your immune system.

A True

B False

A

Quiz time!

Which of these substances are not carried directly in blood plasma?

A Oxygen

B Enzymes

C Carbon dioxide

D Hormones

Quiz time!

Which of these substances are not carried directly in blood plasma?

A Oxygen

B Enzymes

C Carbon dioxide

D Hormones

A